HealthConnect

System Requirements Specification

1. Functional Requirements

Functional requirements describe **what the system MUST do** — the actual features and behaviors.

1.1 User Authentication & Access Control

The system must ensure that only authorized users can access specific features.

Requirements:

- The system shall allow users to log in using a username and password.
- The system shall support role-based access (Admin, Doctor, Nurse, Receptionist).
- The system shall restrict access based on user roles.
- The system shall allow administrators to create, update, and deactivate user accounts.
- The system shall log all login attempts (successful and failed).

Why it matters: Keeps patient data secure and ensures each user sees only what they need.

1.2 Patient Registration & Management

The system must handle patient data efficiently and accurately.

Requirements:

- The system shall allow receptionists to register new patients.
- The system shall collect:
  - Name, age, gender
  - Parent/guardian details
  - Contact information
  - Initial medical condition
- The system shall generate a unique patient ID automatically.
- The system shall prevent duplicate patient entries.
- The system shall allow updating and deleting patient records.
- The system shall allow searching patients by name, ID, or phone number.

Why it matters: Everything in the hospital starts with correct patient data.

1.3 Appointment & Doctor Assignment

The system must organize patient flow and doctor workload.

Requirements:

- The system shall display available doctors and their schedules.
- The system shall assign patients to doctors based on availability.
- The system shall allow rescheduling or reassigning doctors.
- The system shall prevent double booking.
- The system shall maintain appointment history.

Why it matters: Reduces chaos and improves efficiency in daily hospital operations.

1.4 Medical Records Management

The system must maintain complete and accurate patient medical records.

Requirements:

- The system shall allow doctors to:
  - Add diagnosis
  - Prescribe treatment
  - Update patient condition
- The system shall allow nurses to:
  - Record vitals (temperature, blood pressure, etc.)
  - Add observations
- The system shall maintain a history of all updates.
- The system shall allow viewing full patient history.

Why it matters: Doctors need full context to make safe medical decisions.

1.5 Bed Management

The system must track hospital bed availability in real time.

Requirements:

- The system shall display all beds and their status (available/occupied).
- The system shall allow assigning beds to patients.
- The system shall update bed status automatically after discharge.
- The system shall prevent assigning already occupied beds.

Why it matters: Helps avoid confusion and ensures efficient use of resources.

1.6 Discharge Management

The system must support patient discharge process.

Requirements:

- The system shall allow doctors to generate discharge summaries.
- The summary shall include:
  - Diagnosis
  - Treatment
  - Medications
  - Follow-up instructions
- The system shall allow saving and printing discharge summaries.
- The system shall update patient status to "discharged."

Why it matters: Ensures patients leave with clear medical guidance.

1.7 Reporting & Analytics

The system must support decision-making through reports.

Requirements:

- The system shall generate reports on:
  - Patient admissions
  - Discharges
  - Bed occupancy
  - Doctor workload
- The system shall allow filtering by date and category.
- The system shall allow exporting reports (PDF/Excel).

Why it matters: Helps hospital management make better decisions.

1.8 Notifications & Alerts

The system must keep users informed about important events.

Requirements:

- The system shall notify doctors of assigned patients.
- The system shall alert staff for urgent patient conditions.
- The system shall send appointment reminders.
- The system shall notify when beds become available.

Why it matters: Improves response time and patient care.

2. Non-Functional Requirements

Non-functional requirements describe **HOW the system should perform**.

2.1 Performance

- The system shall respond to user actions within **2 seconds**.
- The system shall handle multiple users simultaneously without delay.
- Data retrieval must be fast even with large patient records.
- *Goal:* Smooth and fast experience.

2.2 Security

- The system shall encrypt sensitive patient data.
- The system shall enforce strong passwords.
- The system shall include role-based access control.
- The system shall maintain audit logs for all actions.

Goal: Protect sensitive medical information.

2.3 Usability

- The system shall have a simple and intuitive interface.
- Users should be able to learn the system quickly.
- The system shall minimize user errors (clear forms, validation).

Goal: Easy for doctors and staff to use even under pressure.

2.4 Reliability

- The system shall be available **24/7**.
- The system shall not lose data during failures.
- The system shall include backup and recovery mechanisms.

Goal: Always working, always safe.

2.5 Scalability

- The system shall support increasing numbers of patients and users.
- The system shall handle future expansion (more departments).

Goal: Grow with the hospital.

2.6 Maintainability

- The system shall be easy to update and maintain.
- Code should be modular and well-structured.
- Bugs should be easy to fix.

Goal: Long-term sustainability.

2.7 Compatibility

- The system shall work on different devices (PCs, tablets).
- The system shall be compatible with modern browsers.

Goal: Accessible anywhere in the hospital.